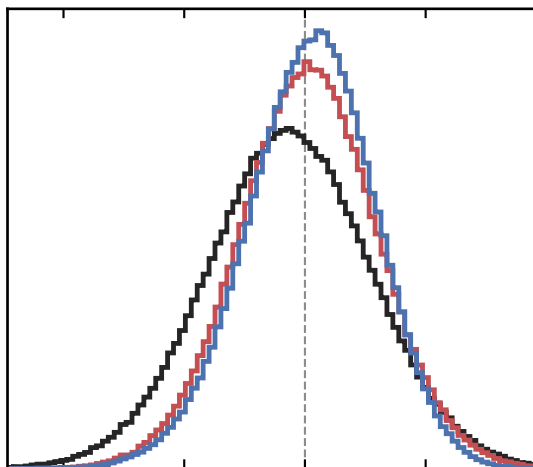
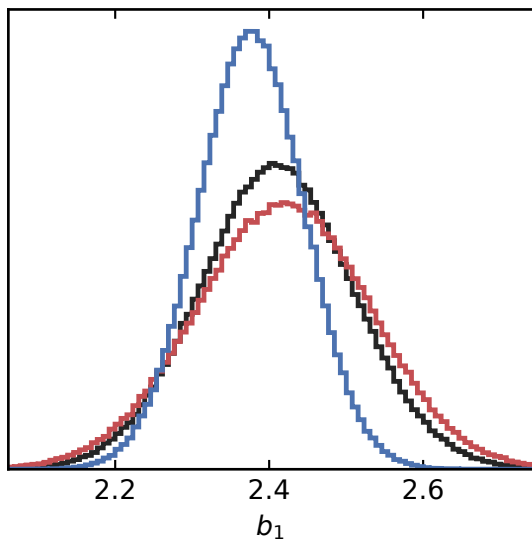
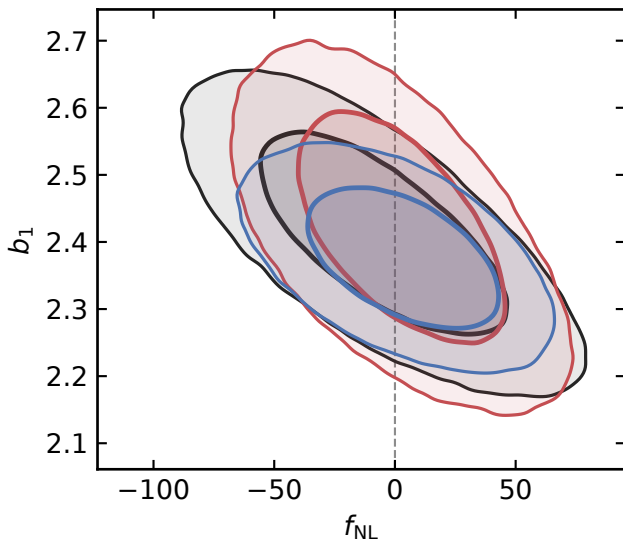


Hybrid + full GIC/RIC | EZmock1000 | Baseline
 $0.4 < z_{\text{obs}} < 0.8$; $k_{\text{max}}=0.08$; $50 \leq s < 350$; BAO mask [80,120]

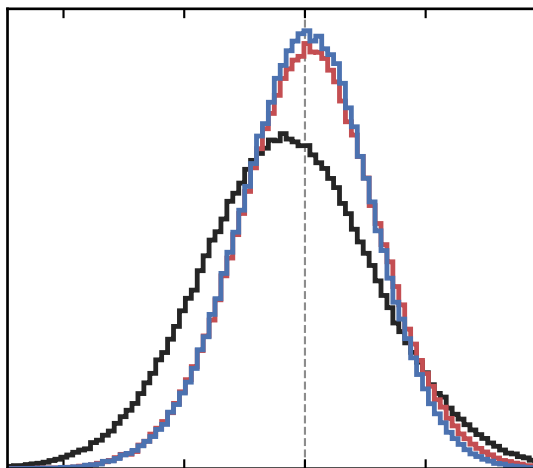


$P_0 + P_2$
 $f_{\text{NL}} = -5.28^{+32.43}_{-34.70}$
 $\xi_0 + \xi_2$
 $f_{\text{NL}} = 3.78^{+25.85}_{-30.57}$
 joint
 $f_{\text{NL}} = 4.72^{+23.56}_{-28.48}$

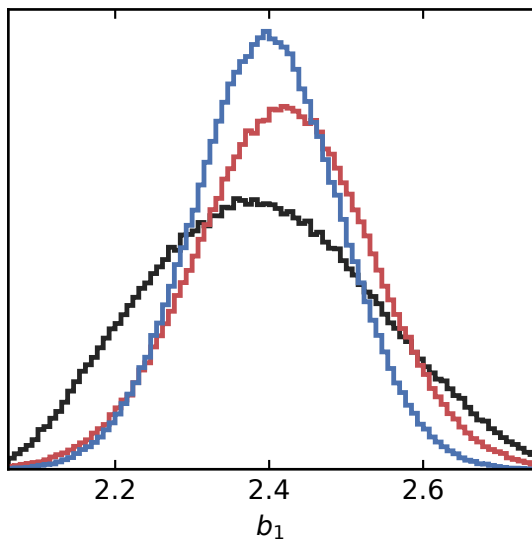
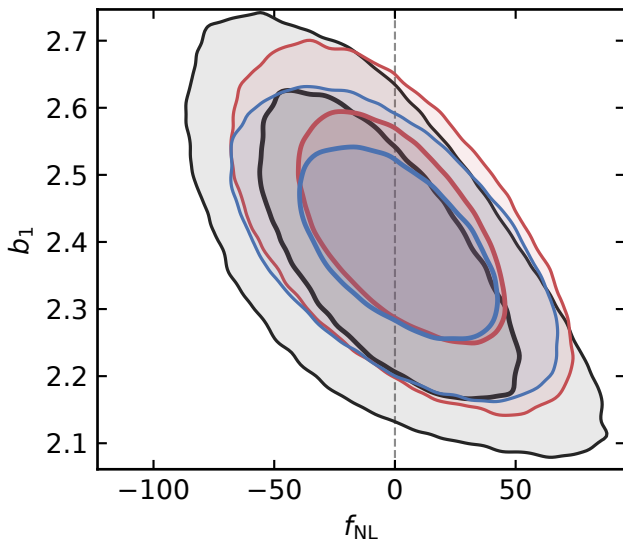


25-halo mean; C_{single} ; Hartlap likelihood; Percival intervals; original 9.18 axes/style

Hybrid + full GIC/RIC | EZmock1000 | $k_{\text{max}} = 0.06$
 $0.4 < z_{\text{obs}} < 0.8$; $k_{\text{max}}=0.06$; $50 \leq s < 350$; BAO mask [80,120)

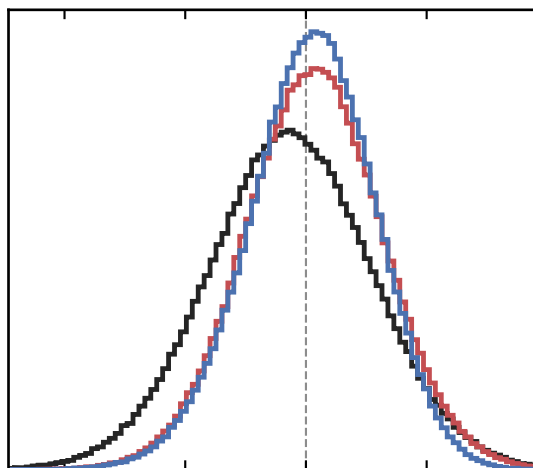


$P_0 + P_2$
 $f_{\text{NL}} = -4.98^{+34.38}_{-37.25}$
 $\xi_0 + \xi_2$
 $f_{\text{NL}} = 3.78^{+25.85}_{-30.57}$
 joint
 $f_{\text{NL}} = 2.49^{+24.64}_{-29.33}$

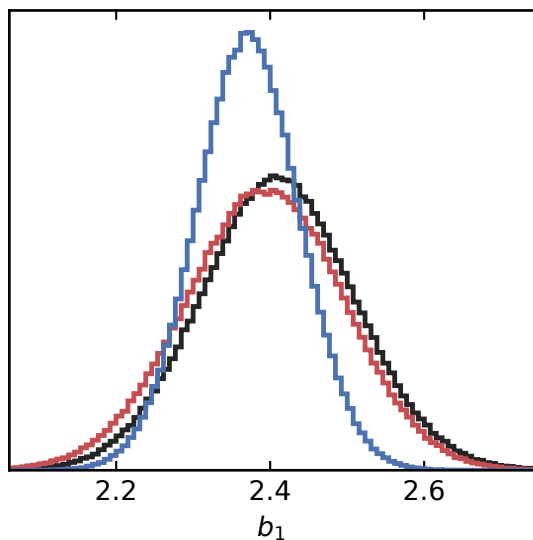
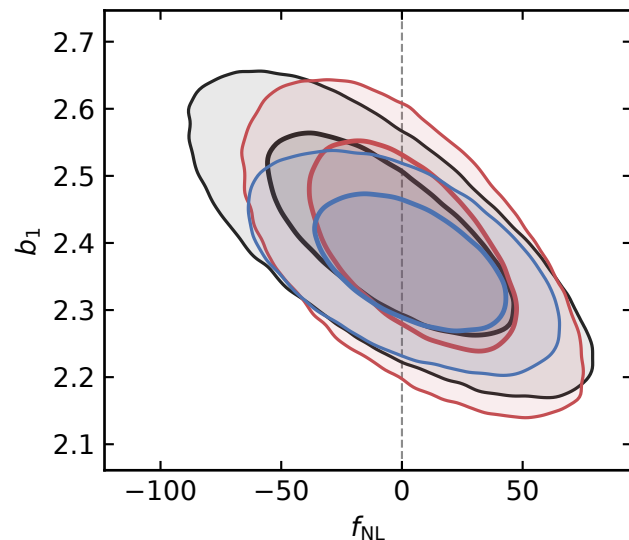


25-halo mean; C_{single} ; Hartlap likelihood; Percival intervals; original 9.18 axes/style

Hybrid + full GIC/RIC | EZmock1000 | BAO unmasked
 $0.4 < z_{\text{obs}} < 0.8$; $k_{\text{max}}=0.08$; $50 \leq s < 350$; no BAO mask

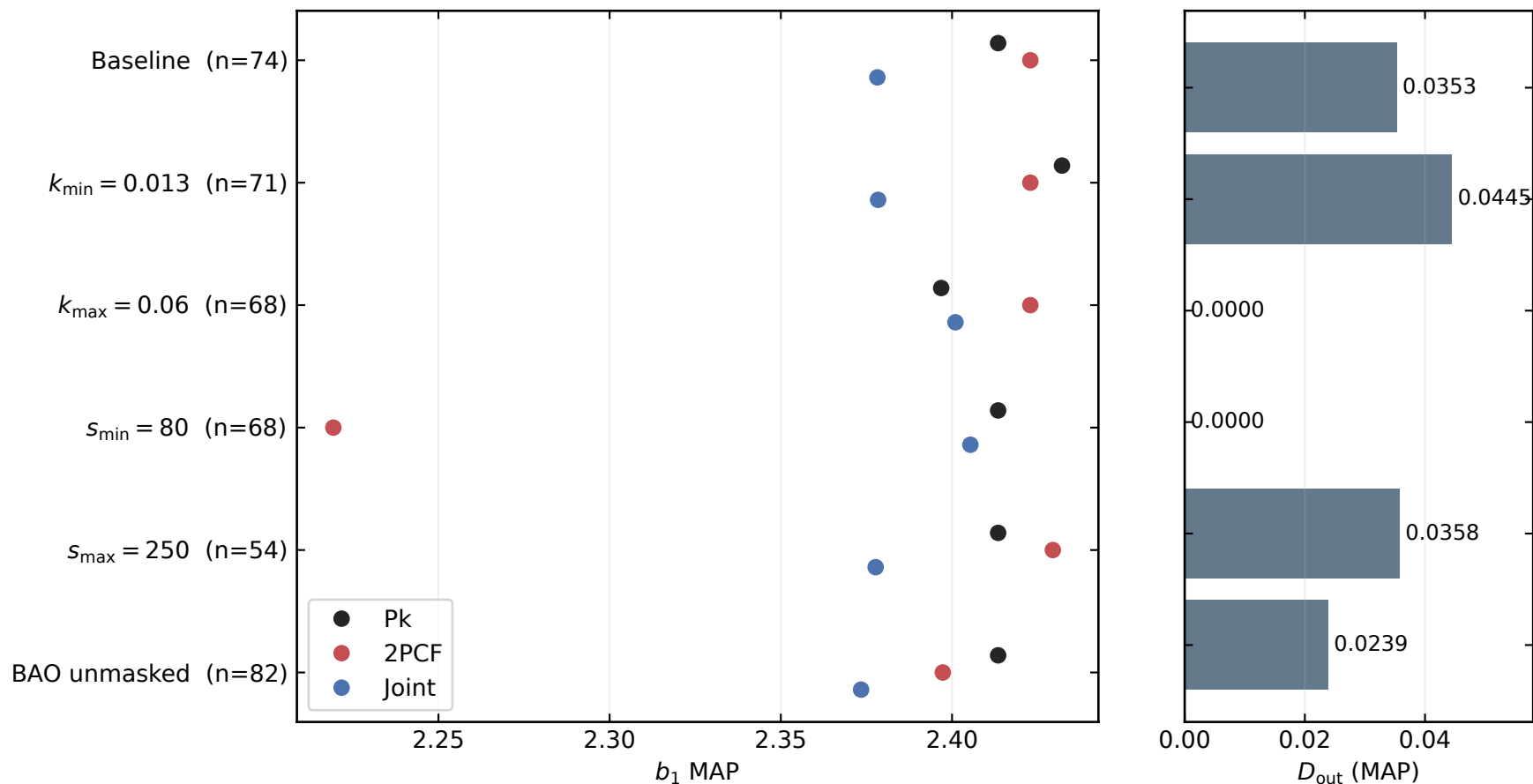


$P_0 + P_2$
 $f_{\text{NL}} = -5.28^{+32.43}_{-34.70}$
 $\xi_0 + \xi_2$
 $f_{\text{NL}} = 5.38^{+25.98}_{-30.20}$
 joint
 $f_{\text{NL}} = 4.57^{+23.52}_{-28.03}$



25-halo mean; C_{single} ; Hartlap likelihood; Percival intervals; original 9.18 axes/style

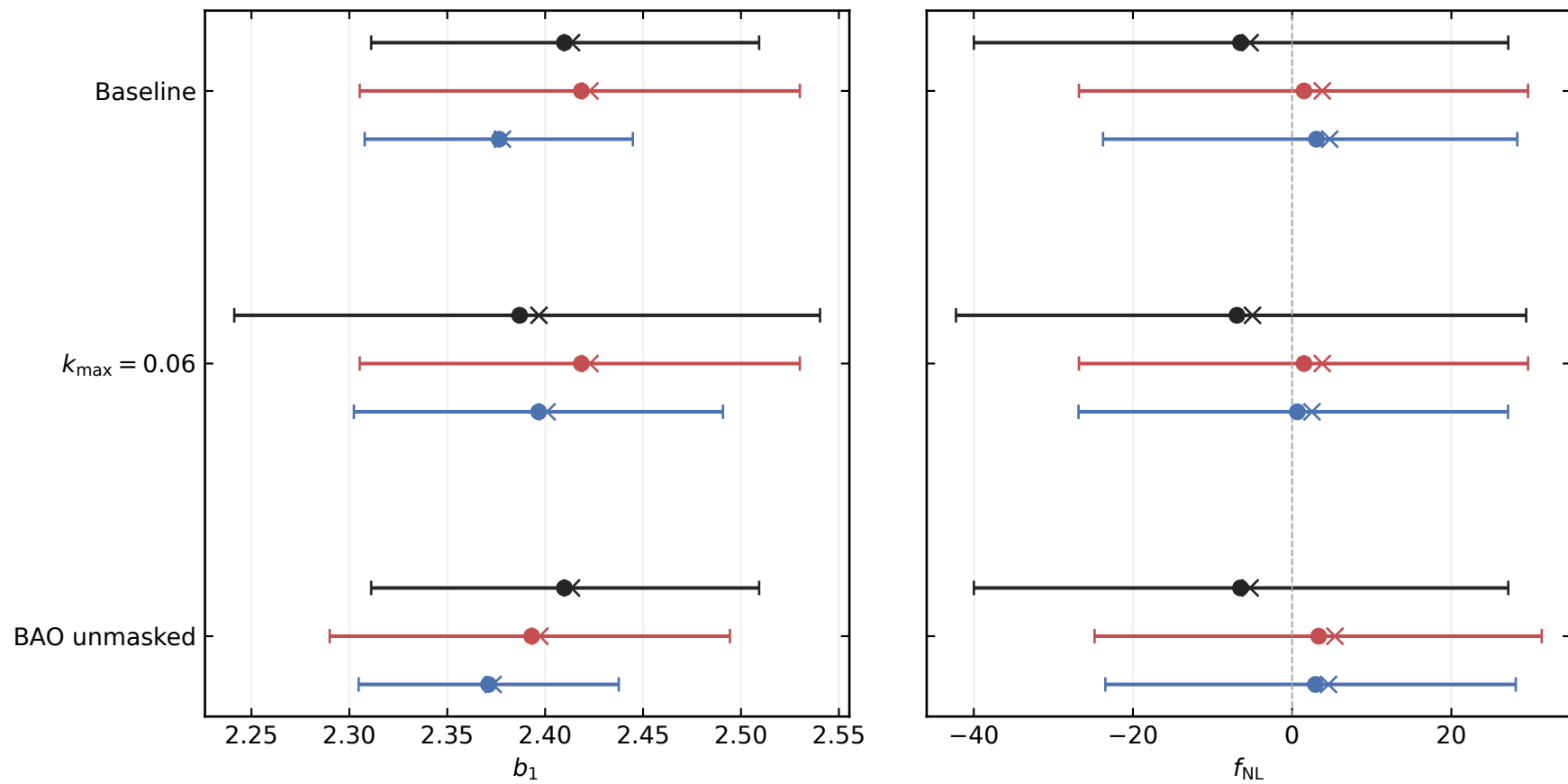
Six fixed selections: baseline + five changes



D_{out} is distance outside the interval between the two single-probe centers, not a tension statistic.
 $s_{\min}=80$ leaves first xi bin at $s=125$ under the BAO mask; xi-only curvature $\sigma(b_1)$ is about 0.54.

Median and 68% interval; crosses mark MAP

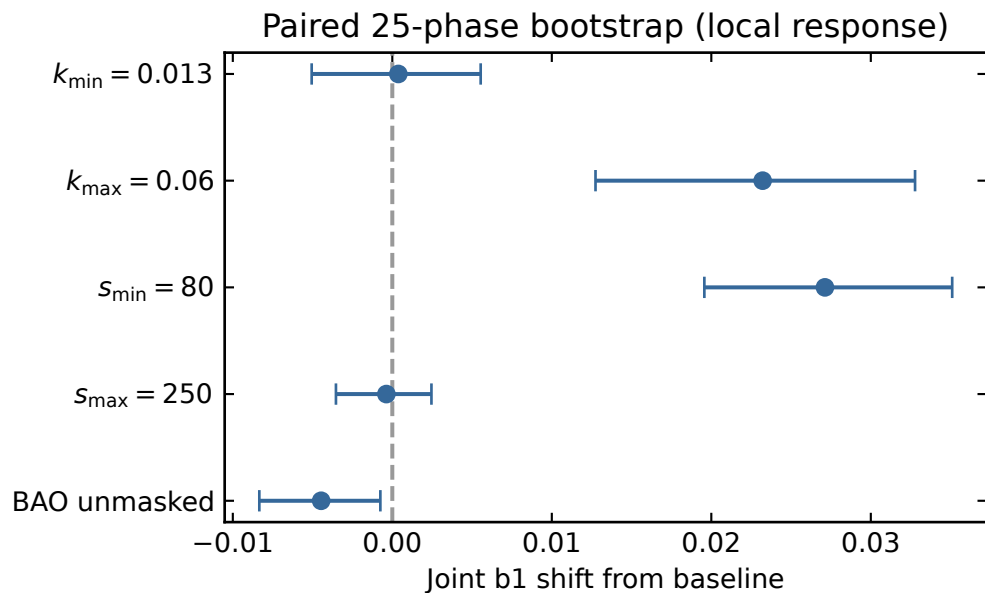
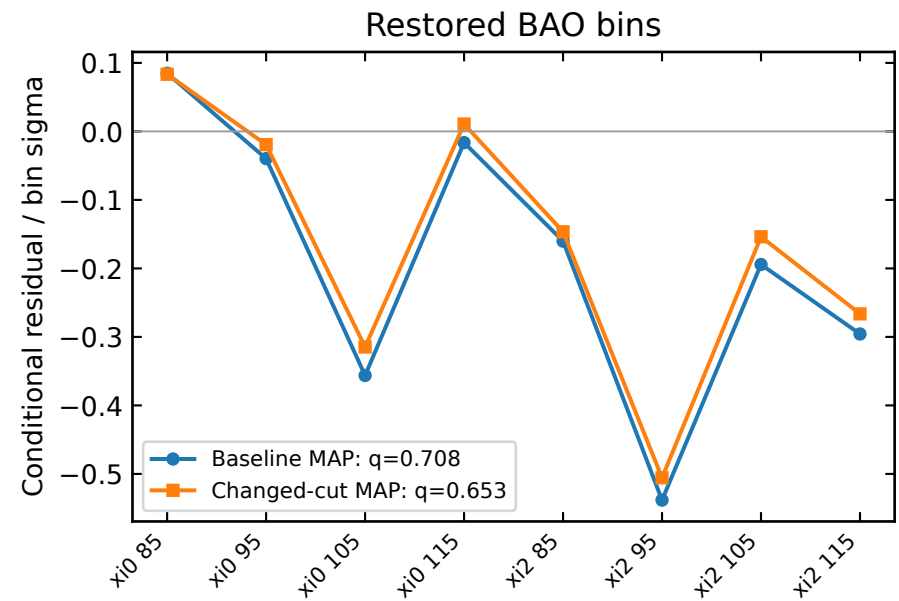
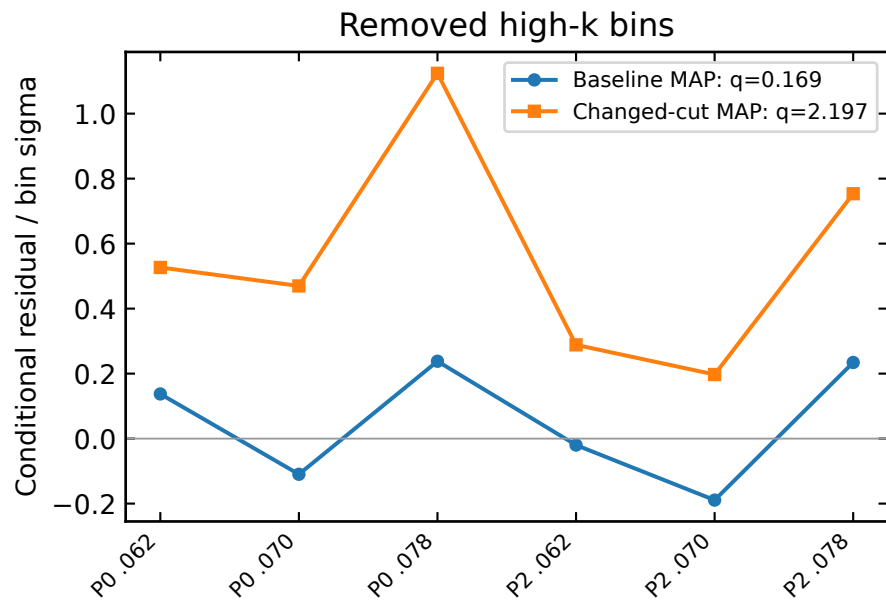
● Pk ● 2PCF ● Joint



$k_{\max}=0.06$ joint width change: b_1 +37.6%; f_{NL} +3.7%.

Four new chains: 64 walkers x 30,000 steps; 5,000 burn-in. Unchanged single-probe chains reused.

Where does the selection dependence come from?



How to interpret these diagnostics

Conditional residuals use the full cross covariance. q is raw C_{single} Schur chi-square at fixed θ . The plotted residual bins remain correlated.

High-k bins are not outliers at the baseline MAP. Changing the cut changes degeneracy directions and information, as well as the central values.

Bootstrap uses a local model and the same 25 phases. It is not a blind validation or a posterior probability. Fit quality cannot be ranked by chi-square values computed on different data vectors.